PREDICT-THEN-OPTIMIZE FRAMEWORK FOR PUBLIC TRANSPORT LINE REDESIGN UNDER FLUCTUATING TRAFFIC CONDITIONS

Zihao Guo, Andrea Araldo*, Fayçal Touzout, Mounim El-Yacoubi

Institut Polytechnique de Paris, Palaiseau 91120, France (Guo, Araldo, El-Yacoubi); National School of State Public Works, Marne-la-Vallée 77454, France (Touzout)

zihao-eric.guo@ip-paris.fr; andrea.araldo@telecom-sudparis.eu;
faycal.touzout@univ-eiffel.fr; mounim.el_yacoubi@telecom-sudparis.eu

* Corresponding author

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Abstract: Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions. This paper proposes a predict-then-optimize method to proactively redesign PT networks under traffic fluctuations, balancing passenger travel time, operating cost, and structural continuity. Forecast inaccuracies are sampled from errors of a deep learning model. Computational results on Mandl's benchmark and the Beijing network show improvements while limiting topological modifications. On the Beijing network under high variability, average travel time improves by up to 25.8% while preserving over 85% line overlap; unlike stop-skipping baselines, the redesign preserves full origin–destination connectivity.

Keywords: Public transport network redesign; Predict-then-optimize; Multi-objective optimization; Travel time uncertainty; Structural continuity

1 Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas et al., 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel & Corman, 2020; Danaher et al., 2020), especially in mixed traffic environments. Such routine perturbations occur daily and can propagate across the network, progressively degrading service reliability (Marra & Corman, 2020).

Preventive strategies enhance robustness ex ante through structural measures, such as dedicated bus lanes that isolate buses from mixed traffic (Russo et al., 2022). However, limited street space and high infrastructure costs often make such measures infeasible in dense urban areas (J.-Q. Li et al., 2009). Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled et al., 2021; Tang et al., 2018; Zheng et al., 2023). A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Network design decisions are revised infrequently, and traditional models assume deterministic or fixed travel times (Büchel & Corman, 2020; Fonseca, 2020), limiting their ability to address recurrent traffic fluctuations.

The main contribution of this paper is to assess the potential of predict-then-optimize proactive redesign of bus networks. To this aim, we propose a new method that adapts bus lines to time-varying stochastic traffic congestion. We solve a multi-objective problem to minimize operational cost and user travel time, and to limit structural network modifications, so as to ease operations and user routing. A possible

deployment rule triggers adaptation only when forecasted conditions sufficiently deviate from nominal conditions; a redesign can be announced in advance (e.g., 1h) to avoid disrupting current journeys. The experiments evaluate a single forecast-conditioned redesign; rolling operation and the trigger are not empirically evaluated.

2 Methodology

2.1 Predict-then-optimize framework

Our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. The prediction model is first trained offline on historical data to learn travel times on road links. During operation, it receives current observations from the latest time steps and outputs a short-term prediction of link travel times $\hat{t}_{ij}$, which are then supplied to the optimization model for potential network redesign. The resulting network design is then evaluated against realized travel times t_{ij} to assess the actual performance of our adaptive network under realistic conditions. To ensure timely responsiveness, this procedure is repeated periodically, i.e., once per hour. At the start of each time slot, the new network is announced in advance (e.g., 1h), so passengers can plan a route that remains valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot (1h).

One possible monitoring rule compares the forecast with nominal link travel times. Let E be the substrate edge set and t^0_{ij} the nominal time of link (i, j). The mean relative forecast deviation is

$$\bar{\Delta} = \frac{1}{|E|} \sum_{(i,j)\in E} \frac{|\hat{t}_{ij} - t^0_{ij}|}{t^0_{ij}}. \tag{1}$$

If $\bar{\Delta}$ exceeds a chosen tolerance, an operator may solve the redesign problem for the next slot; otherwise the announced network is retained. The tolerance and operator approval are deployment choices. This rule is proposed as a possible operational mechanism and is not tested as a multi-hour policy in the experiments.

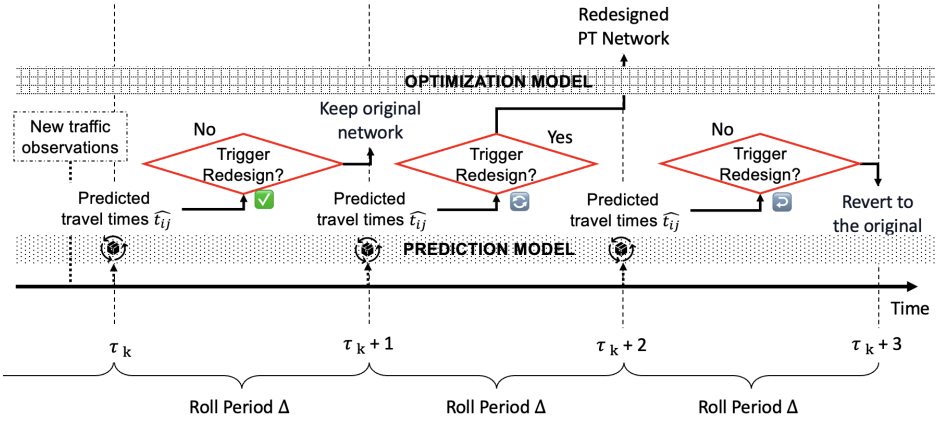


Figure 1: Proactive line redesign framework.

2.2 Network design and objectives

We represent the road infrastructure as a graph with candidate stops and links. Each line is a simple path on this graph and is assigned vehicles subject to a fleet budget. Passengers are assigned to shortest paths on a virtual graph that represents line-specific travel and waiting times. The model requires all considered origin–destination (OD) pairs to remain connected.

Vehicles operating on line ℓ are assumed to traverse its path in both directions, with symmetric travel times. If t^ℓ is its one-way travel time and n^ℓ vehicles are assigned, its headway is $h^\ell = 2t^\ell/n^\ell$. Assuming random

passenger arrivals, average waiting time is $w^\ell = h^\ell/2$. A transfer to another line adds that line's expected waiting time and a fixed five-minute penalty. The virtual graph makes these boarding and transfer costs explicit when evaluating passenger paths.

We formulate our public transport network adaptation problem as a three-objective optimization. The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The use of $\max(\hat{t}_k - t_k^0, 0)$ guarantees that only travel-time increases relative to the nominal benchmark are penalized, thereby avoiding situations where improvements for certain OD pairs compensate for deterioration in others. The second objective Z_2 minimizes a vehicle-time proxy for operational cost. The third objective Z_3 minimizes network structural changes using the Jaccard index, calculated as the ratio between the number of shared edges and the total number of edges. Structural continuity is handled as an objective (Z_3).

$$\min(Z_1, Z_2, Z_3) = \left(\sum_k q_k \max(\hat{t}_k - t_k^0, 0),\ \sum_\ell n^\ell 2\hat{t}^\ell,\ 1 - \frac{\sum_{\ell,\{i,j\}} \min(x_{ij}^\ell, x_{ij}^{\ell,0})}{\sum_{\ell,\{i,j\}} \max(x_{ij}^\ell, x_{ij}^{\ell,0})} \right). \tag{2}$$

Here $\hat{t}_k$ is the predicted shortest travel time for OD pair k on the redesigned network; t_k^0 is its travel time on the original network under nominal link times. For the Jaccard term, x_{ij}^ℓ denotes use of physical link $\{i, j\}$ by line ℓ, and $x_{ij}^{\ell,0}$ its original value. Each line-labelled physical edge is counted once, irrespective of path orientation. Here n^ℓ is the vehicle allocation and $\hat{t}^\ell$ the predicted one-way line travel time; Z_2 is a design-level vehicle-time proxy. When solving our optimization problem, the realized link travel times t_{ij} are not available; we optimize based on predictions $\hat{t}_{ij}$ and then evaluate ex post using realized travel times.

The path and fleet constraints can be expressed compactly as follows. Let N be the stop set, e_{ij}^ℓ indicate the directed path arc from i to j, y^ℓ indicate line activation, a_i^ℓ and b_i^ℓ identify its terminals, and u_i^ℓ be a stop-order variable. Binary variables encode arcs, terminals and activation; n^ℓ is an integer and $\bar{n}$ its per-line upper bound. For each line and candidate stop or arc,

$$\sum_i a_i^\ell = \sum_i b_i^\ell = y^\ell, \qquad a_i^\ell + b_i^\ell \le 1, \tag{3}$$

$$\sum_j e_{ji}^\ell + a_i^\ell = \sum_j e_{ij}^\ell + b_i^\ell, \qquad \sum_j e_{ij}^\ell,\ \sum_j e_{ji}^\ell \le 1, \tag{4}$$

$$u_i^\ell - u_j^\ell + 1 \le |N|(1 - e_{ij}^\ell), \qquad 1 \le u_i^\ell \le |N|, \tag{5}$$

$$e_{ij}^\ell + e_{ji}^\ell \le y^\ell, \qquad y^\ell \le n^\ell \le \bar{n} y^\ell, \quad \sum_\ell n^\ell \le V_{\max}. \tag{6}$$

The first two lines select distinct terminals and enforce flow balance and degree limits. The ordering inequality applies to both candidate directions and eliminates directed subtours. The last line links edge and integer vehicle assignments to line activation and enforces the fleet budget. For each physical link, $x_{ij}^\ell = e_{ij}^\ell + e_{ji}^\ell$; the original-network value is defined analogously.

Passenger routes are not independent network-design variables. For a fixed design, the virtual graph contains line-specific nodes and arcs for travel, access and transfers; waiting time depends on the vehicle allocation. Shortest-path calculations yield the travel time of each OD pair. The model assumes uncongested passenger assignment: it does not model crowding or passenger-flow-dependent travel times.

2.3 Solution algorithm

Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt an NSGA-III framework (Deb & Jain, 2014) to approximate the Pareto front. The evolutionary search operates on network design variables (line topologies e_{ij}^ℓ and fleet allocations n^ℓ). Each individual encodes a complete PT network design, comprising a set of line path representations and the associated vehicle allocations. At each generation, offspring are produced through selection, variation, and evaluation, followed by the

NSGA-III reference-point-based environmental selection, which prioritizes non-dominated solutions while maintaining diversity along the Pareto front. Feasibility-oriented repair addresses line-level structural infeasibility. For each candidate, shortest paths on a virtual-node graph incorporate in-vehicle and waiting times and provide the OD travel times used in the objectives.

Initial networks are constructed from scratch or by locally perturbing the existing routes. Candidates are screened for simple-path validity, connectivity of the considered OD pairs, and a minimum initial Jaccard similarity $J_{\min} = 0.5$. This initialization threshold should not be confused with an enforced overlap threshold for every subsequent solution: structural continuity is an optimization objective.

Crossover exchanges whole lines between two parents, carrying each line's vehicle allocation with its route. Mutation modifies a line locally through node insertion, deletion, or segment reconnection. Feasibility-oriented repair restores invalid line structures and adjusts fleet allocations; it is not guided by objective improvement. The experiments use crossover probability 0.85, mutation probability 0.35, and 125 reference points. The algorithm returns non-dominated alternatives rather than a single universally optimal design.

3 Evaluation method

Our optimization uses predicted link travel times $\hat{t}_{ij}$, while actual link travel times t_{ij} are unknown at decision time. We therefore evaluate whether the generated line plans remain effective in the presence of prediction errors. We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Y. Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead prediction errors. The objective is not to benchmark forecasting models, but to expose the line-plan generation method to realistic forecast errors. Log-relative errors $\eta_{ij} = \log(\hat{t}_{ij}/t_{ij})$ are sampled with replacement from empirical errors; simulated forecasts are $\hat{t}_{ij} = t_{ij}\exp(\eta_{ij})$. Realized link times follow a log-normal model.

Due to the absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Y. Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China. The transfer of forecast-error patterns from Los Angeles to Mandl is an experimental approximation; pooled residual sampling does not represent link-specific heterogeneity or spatio-temporal correlation.

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Ahern et al., 2022; Liang et al., 2025). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. Its origin–destination demand matrix is constructed from smart card data collected during peak hours by aggregating trips between boarding station i and alighting station j.

Key parameters are set as follows: a fixed dwell time of $d = 1$ min per stop is assumed following Chen et al. (2022). For Mandl's network, the maximum fleet size is capped at 40, following the service-frequency setting of Ahern et al. (2022) to maintain comparable headways. For the Beijing network, the maximum is set to 150 to accommodate a constant 12-minute headway across this large PT system. We consider coefficients of variation $cv = 0, 0.05, 0.25, 0.50, 1.00$; the stochastic experiments use 30 realizations of link travel times and prediction errors. The optimization uses predicted times, and designs are evaluated using realized times. A compromise solution is selected from the Pareto front by its Euclidean distance to the component-wise median objective vector.

The Mandl experiments compare redesigned networks with static route-frequency solutions and a frequency-only adjustment, in which line structures remain fixed while vehicle allocation changes.

The Beijing comparison uses the original network and stop-skipping variants. A one-hour maximum computational time per instance is used for the offline experiments; this is a computational budget, not evidence that a complete operational deployment can be carried out within one hour.

We report three complementary indicators. The 20% improvement share (IS_{20}) is the fraction of OD pairs with a travel-time reduction of at least 20% relative to the comparison network. Operational cost reduction (OCR) compares the vehicle-time proxy, and line overlap (LO) measures Jaccard similarity of line-labelled edges. These indicators characterize different trade-offs: a large passenger improvement does not by itself establish either low operating effort or structural continuity.

For the sensitivity analysis, the sampled log-relative residual is multiplied by a factor α, so that $\hat{t}_{ij}(\alpha) = t_{ij}\exp(\alpha\eta_{ij})$. This construction keeps predicted times positive. The same realized link times are used across error scales, separating the effect of forecast quality from changes in the realized traffic scenario. Pareto fronts are evaluated on realized conditions using hypervolume across the performance indicators. This analysis does not compare alternative forecasting architectures.

4 Numerical results

4.1 Mandl benchmark

We use predicted travel times as input to the optimization problem, and we evaluate the resulting adaptive network against realized link travel times and prediction errors across multiple scenarios. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. Across all variability levels and redesign schemes, Line Overlap (LO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring. Relative to the static route-frequency solution of Ahern et al. (2022), for the four-line instance, IS_{20} rises from $5.79 \pm 2.99\%$ at $cv = 0$ to $30.57 \pm 11.60\%$ at $cv = 1$, and OCR rises from $10.20 \pm 5.31\%$ to $39.20 \pm 12.94\%$; line overlap is $84.34 \pm 5.44\%$ and $77.26 \pm 5.19\%$, respectively (mean and standard deviation over 30 realizations).

The frequency-only baseline improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is $16.47 \pm 9.53\%$, compared with $30.57 \pm 11.60\%$ for our method. While frequency-only adjustment keeps the line topology fixed, the proposed line redesign can reroute service away from deteriorated links. In a forecast-error sensitivity analysis, realized times are held fixed while the error scale varies: zero gives a perfect forecast, one reproduces empirical error, and larger values stress the model. The results indicate that the opportunity for redesign grows with variability and that the Pareto front changes little across the tested error scales.

Table 1: Four-line Mandl comparison against the static baseline: mean $\pm$ standard deviation over 30 realizations. IS_{20} is the share of OD pairs with at least 20% improvement.

cv	Method	IS_{20} (%)	OCR (%)	LO (%)
0.05	Frequency only	3.96 ± 2.94	12.85 ± 4.66	100.00 ± 0.00
	Redesign	8.25 ± 4.65	13.28 ± 7.37	83.79 ± 7.73
0.25	Frequency only	8.09 ± 4.35	14.86 ± 5.78	100.00 ± 0.00
	Redesign	15.40 ± 5.66	16.86 ± 6.74	78.83 ± 6.65
0.50	Frequency only	12.94 ± 6.92	19.27 ± 9.04	100.00 ± 0.00
	Redesign	24.63 ± 7.91	24.06 ± 12.01	77.53 ± 5.27
1.00	Frequency only	16.47 ± 9.53	29.40 ± 15.04	100.00 ± 0.00
	Redesign	30.57 ± 11.60	39.20 ± 12.94	77.26 ± 5.19

Table 1 makes the trade-off explicit: frequency-only adjustment preserves all line-labelled edges, whereas redesign changes the network and produces a larger improvement share in the reported comparisons.

The standard deviations also show substantial variation across realizations, so the means should not be interpreted as guarantees for every traffic scenario.

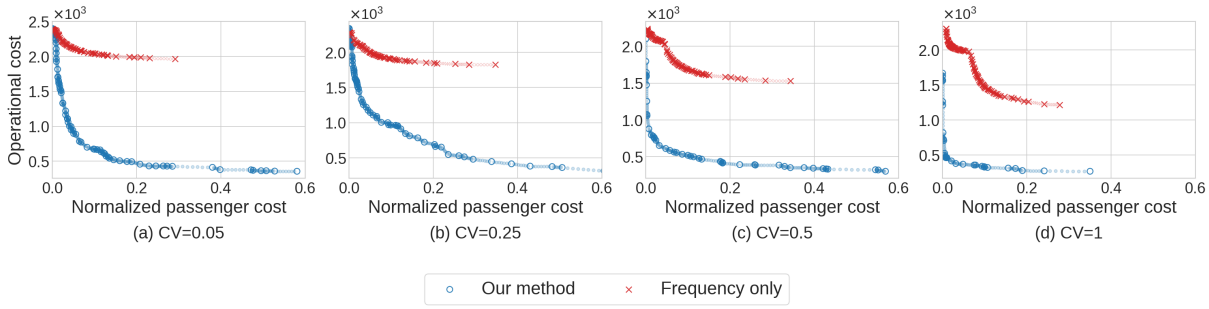


Figure 2: Travel-time and cost (Z_1, Z_2) Pareto-front projections for the four-line Mandl instance under four travel-time variability levels.

Figure 2 plots the two-dimensional projection of the Pareto front, highlighting the trade-off between normalized passenger travel-time deterioration Z_1 and the operating-cost proxy Z_2. Each point maps to a specific network design, with the lower-left corner representing the preferred direction for both minimization objectives. The third objective, structural continuity, is not shown in this projection and is reported as line overlap in Table 1.

4.2 Beijing network

We compare our method against a stop-skipping operational baseline on the Beijing network. For each variability level, links whose predicted travel time exceeds the original by more than a preset threshold are classified as degraded. The stop-skipping strategy then removes intermediate stops adjacent to degraded links whenever a direct shortcut is available, while leaving the overall line structure and vehicle allocation unchanged. This baseline represents a simple, local response to travel-time disruption, but structurally constrained: it cannot reroute vehicles or reassign fleet, and skipping stops may disconnect part of the demand at a stop from the network entirely.

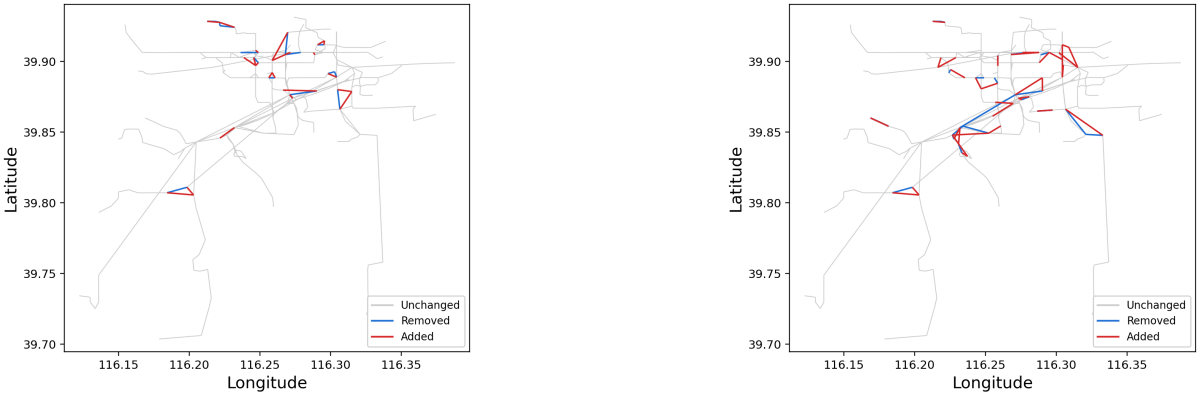


Figure 3: Adapted Beijing networks at nominal and high travel-time variability. Blue indicates a link removed from at least one original line; red indicates a link added to at least one line, as shown in the figure legends.

Figure 3 illustrates the spatial nature of the redesign for two variability settings. A link can be shown as both removed and added when it changes its line assignment, because structural similarity is evaluated using line-labelled edges. The figure illustrates selected designs; the quantitative comparisons below summarize the reported experiments.

The results in Table 3 show that the original network exhibits some inefficiencies: the solution of our optimization yields an 11.40% average travel-time improvement even at $cv = 0$. This reflects static

Table 2: Beijing results at $cv = 1$ compared with the original network. Travel-time improvement for stop-skipping is calculated on reachable OD pairs.

Method	Overlap (%)	Cost reduction (%)	Travel-time improvement (%)	Unreachable OD (%)
Stop-skipping, 25% threshold	42.18	12.71	14.71	38.93
Stop-skipping, 50% threshold	47.83	12.39	19.71	35.28
Stop-skipping, 100% threshold	55.88	13.49	15.11	28.05
Redesign	85.42	22.56	25.78	0

re-optimisation of the incumbent network rather than prediction-driven adaptation. The benefit grows with variability. Table 2 exposes a basic tension in stop-skipping: operational benefit comes at the cost of connectivity. Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows. Because no stops are removed, all OD pairs stay reachable under the redesign. Travel-time improvements for stop-skipping and redesign are calculated on different reachable OD sets and should therefore be interpreted alongside the connectivity figures.

The full improvement at high variability includes both a static redesign effect and adaptation to the scenario, rather than measuring the effect of forecasting alone.

Table 3: Redesign results on the Beijing network across traffic variability levels, compared with the original network.

Coefficient of variation	0	0.05	0.25	0.50	1.00
Line overlap (%)	90.08	87.34	86.51	86.39	85.42
Operational cost reduction (%)	0.50	1.53	3.85	12.02	22.56
Average travel-time improvement (%)	11.40	11.60	16.99	18.44	25.78

Table 3 shows the reported trade-off across all five settings. Average travel-time improvement is computed as the reduction in the sum of OD travel times relative to the original network, divided by the corresponding original sum; it is not the fraction of OD pairs improved by 20%, nor a demand-weighted average. Operational cost reduction reflects the model's vehicle-time proxy, which excludes costs such as crew scheduling and depot operations.

5 Conclusion

This paper systematically studied the potential of proactive bus line redesign, targeted at limited parts of the network, in maintaining user and operator performance in the case of severe traffic fluctuations. We formulated a multi-objective optimization problem balancing passenger travel time, operational costs, and network stability. Our predict-then-optimize approach relies on an NSGA-III-based optimization framework that proactively redesigns bus lines based on predicted line travel times. Experiments on Mandl's network and a real-world Beijing network demonstrate effective adaptation across varying levels of travel-time uncertainty. Overall, the gains grow with travel-time variability and come with high line overlap, indicating that the benefit does not rely on large structural change.

A natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented. A further direction is to incorporate in-vehicle crowding, for instance through a capacitated or congestion-dependent passenger assignment; this would let the redesign account for the load it induces on each line and report crowding-aware rather than purely uncongested travel-time gains.

The operational feasibility of the proposed rolling use is context-dependent. Route changes require advance notice, passenger information, and operator and driver coordination. The numerical experiments

do not measure the costs of communication, retraining, or changing schedules, nor do they validate passenger responses to changing routes. The method can therefore support an operator's assessment of candidate designs, with deployment decisions and practical constraints considered separately.

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